

Tannakian Reconstruction of Program Semantics

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Abstract

We develop a Tannakian reconstruction theory for verified program semantics within the JU_{GEO} framework. The category **Prog** of verified program fragments, equipped with a tensor product given by parallel composition, admits a *neutral Tannakian structure* over the field $k = \mathbb{F}_2$ of truth values. The Tannakian reconstruction theorem identifies the *symmetry group* of a program as the group scheme $\mathcal{G}^{\text{tan}} = \underline{\text{Aut}}^{\otimes}(\omega)$ of tensor-preserving automorphisms of the fiber functor $\omega: \mathbf{Prog} \rightarrow \mathbf{Vect}_k$. The Tannakian fundamental group $\pi_1^{\text{tan}}(\mathbf{Prog})$ classifies all semantics-preserving program transformations: two programs P and Q are semantically equivalent if and only if they have isomorphic Tannakian groups. We apply this characterization to *automated refactoring validation*: the algorithm TANCHECK computes π_1^{tan} before and after a refactoring and certifies semantic equivalence iff the resulting groups are isomorphic. All key theorems are formalized without axioms (beyond the reconstruction principle itself) in the companion LEAN 4 file `Paper15_TannakianReconstruction.lean`.

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1 Introduction

The central question of program verification is when two programs are *the same*—when a refactoring, optimization, or porting preserves observable behavior. The JUGEO framework introduced in Papers 1–14 provides a rich sheaf-theoretic language for expressing and propagating semantic judgments. Yet the *symmetry structure* of those judgments—the group of transformations that leave program semantics unchanged—has received less formal attention.

Tannakian category theory, originating in Grothendieck’s work on motives and later systematized by Deligne and Milne [3], provides exactly the right tool. The classical Tannaka–Krein reconstruction theorem says: given a *neutral Tannakian category* $\mathcal{C}$ with fiber functor $\omega: \mathcal{C} \rightarrow \mathbf{Vect}_k$, the *symmetry group* $G = \underline{\mathrm{Aut}}^\otimes(\omega)$ completely reconstructs $\mathcal{C}$ as $\mathbf{Rep}(G)$.

In this paper we carry out this program for the category **Prog** of JUGEO verified program fragments:

- (i) We define **Prog** as a k -linear rigid symmetric monoidal category whose objects are 8-tuple judgments $\mathcal{J} = (\mathbf{c}, \varphi, \mathcal{A}, \mathcal{E}, \mathcal{O}, \mathcal{B}, \mathcal{T}, \Pi)$ with $\mathcal{T} \preceq \text{SOLVER}$, and whose morphisms are verified program transformations (§3).
- (ii) We construct the *semantic fiber functor* $\omega: \mathbf{Prog} \rightarrow \mathbf{Vect}_k$ that maps each verified fragment to its truth-value vector (§4).
- (iii) We prove that $(\mathbf{Prog}, \omega)$ is a neutral Tannakian category, and state the Tannakian reconstruction theorem identifying $\mathcal{G}^{\text{tan}} = \underline{\mathrm{Aut}}^\otimes(\omega)$ (§5).
- (iv) We define the Tannakian fundamental group $\pi_1^{\text{tan}}(\mathbf{Prog}, P)$ and prove the equivalence criterion: $P \simeq_{\text{tan}} Q$ iff $\pi_1^{\text{tan}}(P) \cong \pi_1^{\text{tan}}(Q)$ (§6).
- (v) We derive the TANCHECK refactoring validation algorithm and prove its soundness and completeness (§7).

Connection to earlier papers. The semantic sites of Paper 1, the descent obstructions of Paper 3, and the characteristic classes of Paper 17 all live naturally in the language of fiber bundles over **Prog**. The present paper adds a Galois-theoretic dimension: just as the étale fundamental group of a scheme classifies its covers, the Tannakian fundamental group of a program classifies its verified transformations.

Organization. Section 2 reviews neutral Tannakian categories. Sections 3–5 construct the main theory. Section 6 defines π_1^{tan} . Section 7 develops the application. Section 8 describes the LEAN 4 formalization. Section 9 discusses related work.

2 Background: Neutral Tannakian Categories

We collect the standard definitions needed for the paper. Throughout, k denotes a field; in our application $k = \mathbb{F}_2$.

Definition 2.1 (Rigid symmetric monoidal category). A *rigid symmetric monoidal category* $(\mathcal{C}, \otimes, \mathbf{1}, \sigma)$ consists of a category $\mathcal{C}$ with a tensor product $\otimes$ that is associative and unital up to coherent isomorphism, a commutativity constraint $\sigma_{X,Y}: X \otimes Y \xrightarrow{\sim} Y \otimes X$, and for each object X a *dual* $X^\vee$ with unit $\eta_X: \mathbf{1} \rightarrow X^\vee \otimes X$ and counit $\varepsilon_X: X \otimes X^\vee \rightarrow \mathbf{1}$ satisfying the zigzag identities.

Definition 2.2 (*k*-linear structure). A *k*-linear category has *k*-vector space structures on all hom-sets, with composition bilinear. An *exact* functor between abelian *k*-linear categories preserves short exact sequences.

Definition 2.3 (Fiber functor and neutral Tannakian category). Let $(\mathcal{C}, \otimes, \mathbf{1})$ be a *k*-linear abelian rigid symmetric monoidal category. A *fiber functor* is an exact faithful *k*-linear tensor functor $\omega: \mathcal{C} \rightarrow \mathbf{Vect}_k$. The pair $(\mathcal{C}, \omega)$ is called a *neutral Tannakian category over k*.

Definition 2.4 (Tannakian group scheme). Given a neutral Tannakian category $(\mathcal{C}, \omega)$, the *Tannakian group scheme* is the functor

$$\mathcal{G}^{\text{tan}}: \mathbf{Alg}_k \longrightarrow \mathbf{Grp}, \quad R \longmapsto \underline{\text{Aut}}^{\otimes}(\omega \otimes_k R),$$

where $\underline{\text{Aut}}^{\otimes}(\omega \otimes R)$ is the group of *R*-linear tensor-compatible automorphisms of $\omega \otimes R$.

The fundamental theorem of Tannakian categories is:

Theorem 2.5 (Grothendieck–Deligne–Milne [3]). *Let $(\mathcal{C}, \omega)$ be a neutral Tannakian category over *k*. Then $\mathcal{G}^{\text{tan}}$ is an affine group scheme over *k*, and the functor*

$$\mathcal{C} \longrightarrow \mathbf{Rep}(\mathcal{G}^{\text{tan}}), \quad X \longmapsto (\omega(X), \text{canonical action of } \mathcal{G}^{\text{tan}})$$

is an equivalence of tensor categories.

Remark 2.6. For a finite group *G*, the category $\mathbf{Rep}_k(G)$ of finite-dimensional *k*-linear *G*-representations is a neutral Tannakian category with fiber functor the forgetful functor, and $\underline{\text{Aut}}^{\otimes}(\text{forget}) \cong G$. This is the motivating example.

3 The Category of Verified Program Fragments

3.1 Objects

Recall from Papers 1–4 that a JU_{GEO} judgment is an 8-tuple

$$\mathcal{J} = (\mathbf{c}, \varphi, \mathcal{A}, \mathcal{E}, \mathcal{O}, \mathcal{B}, \mathcal{T}, \Pi),$$

where *c* is a coordinate, φ is a proposition about the program, $\mathcal{A}$ is the carrier type of the evidence, $\mathcal{E}$ is the evidence bundle, $\mathcal{O}$ encodes remaining proof obligations, $\mathcal{B}$ classifies obstruction cohomology classes (cf. Paper 3), $\mathcal{T}$ is the trust tier, and Π is the provenance chain.

Definition 3.1 (Verified program fragment). A *verified program fragment* is a pair $(P, \mathcal{J})$ consisting of:

- a program fragment *P* with coordinate $\mathbf{c} = \text{pc}(P)$;
- a JU_{GEO} judgment $\mathcal{J}$ such that the trust tier satisfies $\text{SOLVER} \preceq \mathcal{T}$ (i.e. the evidence has been validated by an SMT solver or formal proof).

We write $\text{Ob}(\mathbf{Prog})$ for the class of all verified program fragments.

Requiring $\text{SOLVER} \preceq \mathcal{T}$ means the proposition φ —the semantic specification of *P*—has been discharged by at least an SMT-solver certificate. Higher trust tiers (PROOF) are also allowed.

3.2 Morphisms

Definition 3.2 (Verified transformation). A *verified transformation*

$$T: (P, \mathcal{J}_P) \longrightarrow (Q, \mathcal{J}_Q)$$

is a program transformation $T: P \rightarrow Q$ together with a verification certificate witnessing that T preserves semantics:

$$\omega(\mathcal{J}_P) = \omega(\mathcal{J}_Q) \circ \hat{T}, \quad (1)$$

where $\hat{T}$ is the induced map on semantic vectors (Definition 4.1).

Identity morphisms are the identity transformations; composition is sequential program composition. Preservation of semantics composes because equation (1) is transitive.

Lemma 3.3 (Prog is a category). *The above data define a category **Prog**.*

Proof. Identity preservation: $\omega(\mathcal{J}_P) = \omega(\mathcal{J}_P) \circ \hat{\text{id}}$ because $\hat{\text{id}} = \text{id}$. Composition: if $T: P \rightarrow Q$ and $S: Q \rightarrow R$ satisfy (1), then $\widehat{S \circ T} = \hat{S} \circ \hat{T}$, so $\omega(\mathcal{J}_R) \circ (\hat{S} \circ \hat{T}) = (\omega(\mathcal{J}_R) \circ \hat{S}) \circ \hat{T} = \omega(\mathcal{J}_Q) \circ \hat{T} = \omega(\mathcal{J}_P)$. $\square$

3.3 Tensor Product

Definition 3.4 (Parallel composition). The *tensor product* in **Prog** is parallel composition:

$$(P, \mathcal{J}_P) \otimes (Q, \mathcal{J}_Q) := (P \parallel Q, \mathcal{J}_P \otimes_{\text{sem}} \mathcal{J}_Q),$$

where $P \parallel Q$ is the program that runs P and Q on disjoint inputs, and $\mathcal{J}_P \otimes_{\text{sem}} \mathcal{J}_Q$ is the componentwise join of the two judgments with proposition $\varphi_P \wedge \varphi_Q$ and evidence bundle $\mathcal{E}_P \oplus \mathcal{E}_Q$.

The monoidal unit is the *empty program* $(\varepsilon, \mathcal{J}_\varepsilon)$ with trivial specification $\top$.

Lemma 3.5 (Symmetry). *There is a natural isomorphism $\sigma_{P,Q}: P \otimes Q \xrightarrow{\sim} Q \otimes P$ given by swapping the two parallel components.*

3.4 Duals and Rigidity

Definition 3.6 (Dual fragment). The dual $(P, \mathcal{J}_P)^\vee$ is the program $P^\vee$ whose input and output roles are swapped, with evaluation functor $\omega(P^\vee) := \omega(P)^* = \text{Hom}_k(\omega(P), k)$.

The unit $\eta_P: \mathbf{1} \rightarrow P^\vee \otimes P$ and counit $\varepsilon_P: P \otimes P^\vee \rightarrow \mathbf{1}$ are given by the standard evaluation and coevaluation maps in **Vect**_k.

Proposition 3.7 (Prog is rigid). *With the above duals, **Prog** is a rigid symmetric monoidal k -linear category.*

Remark 3.8. In practice, duals correspond to the *inverse specification*: if P transforms input type A to output type B , then $P^\vee$ is any right-inverse satisfying the adjunction equation. The existence of such a dual is guaranteed within the verified fragment model whenever φ_P encodes a bijective input–output relation. For the refactoring application (§7) we only need the tensor structure, not duals explicitly.

4 The Semantic Fiber Functor

4.1 Construction

We fix a countable set of *test cases* $\mathcal{T} = \{t_1, t_2, \dots\}$ and define the base field $k = \mathbb{F}_2$ with truth values $\{0, 1\}$.

Definition 4.1 (Semantic fiber functor). Let $(P, \mathcal{J})$ be a verified program fragment with specification φ containing n test cases. The *semantic fiber functor* assigns

$$\omega(P, \mathcal{J}) := \bigoplus_{i=1}^n k \cdot e_i \cong k^n,$$

where $e_i = 1$ if P satisfies test case t_i and $e_i = 0$ otherwise. On morphisms, a verified transformation $T: (P, \mathcal{J}_P) \rightarrow (Q, \mathcal{J}_Q)$ induces the linear map $\omega(T) = \text{id}_{k^n}$ (since T preserves semantics by (1)).

We may write $\llbracket P \rrbracket$ for $\omega(P, \mathcal{J}_P)$ when the judgment is clear from context.

Theorem 4.2 (Fiber functor properties). *The functor $\omega: \mathbf{Prog} \rightarrow \mathbf{Vect}_k$ is:*

- (a) **Faithful**: $\omega(T) = \omega(S)$ whenever T, S are verified transformations with the same source and target;
- (b) **Exact**: ω maps short exact sequences in $\mathbf{Prog}$ to short exact sequences in $\mathbf{Vect}_k$;
- (c) **Tensor**: there are natural isomorphisms $\omega(P \otimes Q) \cong \omega(P) \otimes \omega(Q)$ and $\omega(\mathbf{1}) \cong k$.

Proof. (a) In $\mathbf{Prog}$, two morphisms $T, S: P \rightarrow Q$ are equal iff they induce the same semantic map, which is precisely $\omega(T) = \omega(S)$ by definition. (b) A short exact sequence $0 \rightarrow A \rightarrow B \rightarrow C \rightarrow 0$ in $\mathbf{Prog}$ corresponds to a semantic filtration; ω sends it to the corresponding filtration of k -vector spaces, which is exact because k is a field. (c) By Definition 3.4, $\omega(P \otimes Q) = k^{n_P + n_Q} \cong k^{n_P} \otimes k^{n_Q} = \omega(P) \otimes \omega(Q)$. The unit isomorphism $\omega(\mathbf{1}) \cong k^0 = k$ is immediate from the empty specification. $\square$

4.2 Diagrams

The following commutative diagram summarizes the relationship between $\mathbf{Prog}$ and $\mathbf{Vect}_k$ via ω :

$$\begin{array}{ccc} \mathbf{Prog} \otimes \mathbf{Prog} & \xrightarrow{\omega \times \omega} & \mathbf{Vect}_k \times \mathbf{Vect}_k \\ \otimes_{\mathbf{Prog}} \downarrow & & \downarrow \otimes_{\mathbf{Vect}_k} \\ \mathbf{Prog} & \xrightarrow{\omega} & \mathbf{Vect}_k \end{array}$$

The commutativity of this diagram is part (c) of Theorem 4.2.

5 Neutral Tannakian Structure and Reconstruction

5.1 Main Structure Theorem

Theorem 5.1 (Prog is a neutral Tannakian category). *The pair $(\mathbf{Prog}, \omega)$ is a neutral Tannakian category over $k = \mathbb{F}_2$. In particular:*

- (i) **Prog** is k -linear, abelian, and semisimple;
- (ii) **Prog** is rigid symmetric monoidal (Proposition 3.7);
- (iii) ω is a faithful exact tensor functor (Theorem 4.2).

Proof sketch. Linearity: $\text{Hom}_{\mathbf{Prog}}((P, \mathcal{J}_P), (Q, \mathcal{J}_Q))$ is a k -vector space because verified transformations can be combined linearly over $k = \mathbb{F}_2$ (XOR of program outputs). Abelian: kernels and cokernels exist as sub-specifications and quotient specifications. Semisimplicity: Every verified fragment is a direct sum of *atomic* fragments—those whose specification φ is irreducible—because the Boolean algebra of specifications is semisimple over $\mathbb{F}_2$. $\square$

Remark 5.2. The semisimplicity assumption is the key idealization. In the finite-test-case model, every object of **Prog** decomposes as a direct sum of rank-1 objects (k, e_i) —the individual test cases. This mirrors the semisimplicity of representations of a finite group over an algebraically closed field of characteristic 0 (or characteristic $p \nmid |G|$).

5.2 The Tannakian Group Scheme

Definition 5.3 (Tannakian group of a program). The *Tannakian group scheme* of **Prog** is the affine k -group scheme

$$\mathcal{G}^{\text{tan}} := \underline{\text{Aut}}^{\otimes}(\omega),$$

defined on k -algebras R by

$$\mathcal{G}^{\text{tan}}(R) = \{ \alpha = (\alpha_X)_{X \in \text{Ob}(\mathbf{Prog})} \mid \alpha_X \in \text{GL}_R(\omega(X) \otimes R), \alpha \text{ is natural and tensor-compatible} \}.$$

Here *tensor-compatible* means that for all $X, Y \in \text{Ob}(\mathbf{Prog})$,

$$\alpha_{X \otimes Y} = \alpha_X \otimes \alpha_Y \quad \text{and} \quad \alpha_{\mathbf{1}} = \text{id}_R.$$

Theorem 5.4 (Reconstruction). *Let $(\mathbf{Prog}, \omega)$ be as in Theorem 5.1. Then:*

- (i) $\mathcal{G}^{\text{tan}}$ is an affine group scheme over k ;
- (ii) there is an equivalence of k -linear tensor categories

$$\mathbf{Prog} \xrightarrow{\sim} \mathbf{Rep}(\mathcal{G}^{\text{tan}});$$

- (iii) two objects $(P, \mathcal{J}_P), (Q, \mathcal{J}_Q) \in \mathbf{Prog}$ are isomorphic if and only if $\omega(P, \mathcal{J}_P) \cong \omega(Q, \mathcal{J}_Q)$ as $\mathcal{G}^{\text{tan}}$ -representations.

Proof. Part (i) follows from the general Tannakian theory (Theorem 2.5) applied to $(\mathbf{Prog}, \omega)$. Part (ii) is the content of the Grothendieck–Deligne–Milne reconstruction theorem. The key step is the construction of the *reconstruction morphism*

$$\varrho: \mathbf{Prog} \longrightarrow \mathbf{Rep}(\mathcal{G}^{\text{tan}}), \quad (P, \mathcal{J}) \longmapsto (\omega(P, \mathcal{J}), \rho_{(P, \mathcal{J})}),$$

where $\rho_{(P, \mathcal{J})}(g) = g_{(P, \mathcal{J})}$ is the tautological $\mathcal{G}^{\text{tan}}$ -action. Faithfulness of ϱ follows from faithfulness of ω ; fullness from the fact that every $\mathcal{G}^{\text{tan}}$ -equivariant map between $\omega(P)$ and $\omega(Q)$ lifts to a morphism in **Prog**. Part (iii) is a special case of the general equivalence: under ϱ , isomorphic objects in **Prog** correspond to isomorphic $\mathcal{G}^{\text{tan}}$ -representations. $\square$

5.3 The Reconstruction Triangle

The full reconstruction picture is captured by the following diagram:

$$\begin{array}{ccc}
 \mathbf{Prog} & \xrightarrow[\sim]{\varrho} & \mathbf{Rep}(\mathcal{G}^{\tan}) \\
 & \searrow \omega & \swarrow \text{forget} \\
 & \mathbf{Vect}_k &
 \end{array}$$

The fiber functor ω factors through the forgetful functor from $\mathbf{Rep}(\mathcal{G}^{\tan})$ to $\mathbf{Vect}_k$, and ϱ is the equivalence making the triangle commute up to natural isomorphism.

6 The Tannakian Fundamental Group

6.1 Definition and Basic Properties

Definition 6.1 (Tannakian fundamental group). Let $P_0 = (P_0, \mathcal{J}_0) \in \text{Ob}(\mathbf{Prog})$ be a basepoint. The *Tannakian fundamental group* of $\mathbf{Prog}$ at P_0 is

$$\pi_1^{\tan}(\mathbf{Prog}, P_0) := \underline{\text{Aut}}^{\otimes}(\omega|_{\langle P_0 \rangle}),$$

where $\langle P_0 \rangle$ denotes the full Tannakian subcategory generated by P_0 , and $\omega|_{\langle P_0 \rangle}$ is the restriction of ω to this subcategory.

Concretely, $\pi_1^{\tan}(\mathbf{Prog}, P_0)$ is the group of all tensor-compatible automorphisms of the vector $\omega(P_0) \in \mathbf{Vect}_k$.

Proposition 6.2 (Group structure). $\pi_1^{\tan}(\mathbf{Prog}, P_0)$ is a group under composition of natural automorphisms. The group operation is

$$(\alpha \cdot \beta)_X := \alpha_X \circ \beta_X \quad \text{for all } X \in \langle P_0 \rangle,$$

with identity the collection of identity maps $\text{id}_{\omega(X)}$.

Proof. Closure: if α and β are tensor-compatible, so is $\alpha \cdot \beta$ (since automorphisms compose and tensor compatibility is preserved). Identity: id is clearly tensor-compatible. Inverses: α^{-1} exists because each α_X is a linear automorphism, and $(\alpha^{-1})_{X \otimes Y} = \alpha_{X \otimes Y}^{-1} = (\alpha_X \otimes \alpha_Y)^{-1} = \alpha_X^{-1} \otimes \alpha_Y^{-1}$. Associativity: inherited from composition of functions. $\square$

6.2 Change of Basepoint

Proposition 6.3 (Basepoint independence). For any two basepoints $P_0, P_1 \in \text{Ob}(\mathbf{Prog})$ with $\omega(P_0) \cong \omega(P_1)$, there is a (non-canonical) isomorphism

$$\pi_1^{\tan}(\mathbf{Prog}, P_0) \cong \pi_1^{\tan}(\mathbf{Prog}, P_1).$$

Proof. An isomorphism $\phi: \omega(P_0) \xrightarrow{\sim} \omega(P_1)$ induces a conjugation isomorphism $\alpha \mapsto \phi \circ \alpha \circ \phi^{-1}$. $\square$

6.3 The Equivalence Criterion

The following theorem is the central result of the paper:

Theorem 6.4 (Tannakian Equivalence Criterion). *Two verified program fragments $(P, \mathcal{J}_P)$ and $(Q, \mathcal{J}_Q)$ are semantically equivalent—that is, $\omega(P, \mathcal{J}_P) \cong \omega(Q, \mathcal{J}_Q)$ as objects of $\mathbf{Vect}_k$ —if and only if*

$$\pi_1^{\text{tan}}(\mathbf{Prog}, P) \cong \pi_1^{\text{tan}}(\mathbf{Prog}, Q).$$

Proof. ($\Rightarrow$): If $\omega(P) \cong \omega(Q)$, then by Proposition 6.3 the Tannakian fundamental groups are isomorphic.

($\Leftarrow$): If $\pi_1^{\text{tan}}(P) \cong \pi_1^{\text{tan}}(Q)$, then by the reconstruction theorem (Theorem 5.4(iii)), $(P, \mathcal{J}_P)$ and $(Q, \mathcal{J}_Q)$ correspond to isomorphic $\mathcal{G}^{\text{tan}}$ -representations in $\mathbf{Rep}(\mathcal{G}^{\text{tan}})$, hence $\omega(P) \cong \omega(Q)$ in $\mathbf{Vect}_k$. $\square$

Corollary 6.5 (Semantic invariance). *Semantic equivalence is completely captured by the Tannakian fundamental group: no information about the program is lost by passing to π_1^{tan} .*

6.4 Comparison with Topological π_1

The analogy between π_1^{tan} and the classical topological fundamental group π_1 is more than formal:

Topology	Galois Theory	Program Semantics
Space X	Scheme S	Category Prog
Path space	Étale cover	Verified transformation
$\pi_1(X, x_0)$	$\pi_1^{\text{ét}}(S, s_0)$	$\pi_1^{\text{tan}}(\mathbf{Prog}, P_0)$
Covering spaces	Galois extensions	Verified extensions
Deck transformations	Galois group	Semantics-preserving refactorings

Remark 6.6. In Grothendieck’s original vision, the étale fundamental group $\pi_1^{\text{ét}}(S, \bar{s})$ classifies the finite étale covers of S . By analogy, $\pi_1^{\text{tan}}(\mathbf{Prog}, P_0)$ classifies the verified “covers” of P_0 —the verified sub-programs whose semantics ω includes as a direct summand.

7 Application: Automated Refactoring Validation

7.1 The Refactoring Problem

A *refactoring* is a program transformation $R: P \rightarrow P'$ that is claimed to preserve program behavior. In the JUGE model, a refactoring is a pair of verified fragments $(P, \mathcal{J}_P)$ and $(P', \mathcal{J}_{P'})$ together with an assertion of semantic equivalence. The fundamental question is: how do we certify this equivalence efficiently?

Traditional approaches require either:

- (a) full equivalence checking (undecidable in general), or
- (b) running a fixed test suite (incomplete: misses untested behaviors).

The Tannakian approach provides a principled middle ground: compute π_1^{tan} before and after the refactoring, and check whether the resulting groups are isomorphic.

7.2 The TanCheck Algorithm

Definition 7.1 (TANCHECK). Given two verified fragments $(P, \mathcal{J}_P)$ and $(P', \mathcal{J}_{P'})$ sharing the same specification type, the TANCHECK *algorithm* proceeds:

1. Compute the semantic vectors $v = \omega(P, \mathcal{J}_P) \in k^n$ and $v' = \omega(P', \mathcal{J}_{P'}) \in k^n$.
2. Check whether $v = v'$ (as elements of k^n).
3. Output EQUIVALENT if $v = v'$, otherwise output NON-EQUIVALENT with a counterexample test case t_i such that $v_i \neq v'_i$.

Theorem 7.2 (Soundness of TANCHECK). *If TANCHECK(P, P') outputs EQUIVALENT, then $\pi_1^{\text{tan}}(\mathbf{Prog}, P) \cong \pi_1^{\text{tan}}(\mathbf{Prog}, P')$.*

Proof. If $v = v'$ then $\omega(P) \cong \omega(P')$ in $\mathbf{Vect}_k$ (they are equal as vectors). By Theorem 6.4, $\pi_1^{\text{tan}}(P) \cong \pi_1^{\text{tan}}(P')$. $\square$

Theorem 7.3 (Completeness of TANCHECK). *If $\pi_1^{\text{tan}}(\mathbf{Prog}, P) \cong \pi_1^{\text{tan}}(\mathbf{Prog}, P')$, then TANCHECK(P, P') outputs EQUIVALENT.*

Proof. By Theorem 6.4, $\pi_1^{\text{tan}}(P) \cong \pi_1^{\text{tan}}(P')$ implies $\omega(P) \cong \omega(P')$, i.e. $v = v'$, so TANCHECK outputs EQUIVALENT. $\square$

Corollary 7.4 (Correctness). TANCHECK(P, P') = EQUIVALENT *if and only if* $\pi_1^{\text{tan}}(\mathbf{Prog}, P) \cong \pi_1^{\text{tan}}(\mathbf{Prog}, P')$.

7.3 Worked Example: Loop Unrolling

Consider a simple loop:

Listing 1: Original: bounded loop

```
1 def sum_first_n(n: int) -> int:
2     """Return 1 + 2 + ... + n."""
3     acc = 0
4     for i in range(1, n + 1):
5         acc += i
6     return acc
```

and its refactoring to the closed-form formula:

Listing 2: Refactored: closed form

```
1 def sum_first_n(n: int) -> int:
2     """Return 1 + 2 + ... + n."""
3     return n * (n + 1) // 2
```

The specification is $\varphi = \{\text{sum_first_n}(n) = n(n+1)/2 \mid n \in \{0, 1, \dots, 9\}\}$, giving $n = 10$ test cases.

Applying TanCheck.

- $\omega(\text{sum_loop}) = (1, 1, 1, 1, 1, 1, 1, 1, 1, 1) \in \mathbb{F}_2^{10}$ (all tests pass).
- $\omega(\text{sum_closed}) = (1, 1, 1, 1, 1, 1, 1, 1, 1, 1) \in \mathbb{F}_2^{10}$ (all tests pass).
- $v = v'$, so TANCHECK outputs EQUIVALENT.
- Conclusion: $\pi_1^{\text{tan}}(\text{sum_loop}) \cong \pi_1^{\text{tan}}(\text{sum_closed})$, and the refactoring is certified.

Tannakian group. Both programs generate the same subcategory of **Prog**, isomorphic to $\mathbf{Rep}(\mathbb{F}_2^\times)$ where $\mathbb{F}_2^\times = \{1\}$ is the trivial group. The Tannakian fundamental group is thus trivial, reflecting that there are no non-trivial semantic symmetries (the specification is completely rigid).

7.4 Worked Example: Bug Detection

Consider a buggy refactoring:

Listing 3: Buggy refactoring (off-by-one)

```
1 def sum_first_n(n: int) -> int:
2     """Return 1 + 2 + ... + n."""
3     return n * (n - 1) // 2    # BUG: should be (n+1)
```

Applying TanCheck.

- $\omega(\text{sum_loop}) = (1, 1, 1, 1, 1, 1, 1, 1, 1, 1)$.
- $\omega(\text{sum_buggy}) = (1, 0, 0, 0, 0, 0, 0, 0, 0, 0)$ (only $n = 0$ passes: $0 = 0$).
- $v \neq v'$, so TANCHECK outputs NON-EQUIVALENT with counterexample t_2 ($n = 1$, expected 1, got 0).
- Conclusion: $\pi_1^{\text{tan}}(\text{sum_loop}) \not\cong \pi_1^{\text{tan}}(\text{sum_buggy})$, and the refactoring is rejected.

7.5 Complexity Analysis

Proposition 7.5 (Complexity of TANCHECK). *Given verified fragments with n test cases each, TANCHECK runs in time $O(n)$: it simply computes ω (evaluates each test case in constant time) and compares two k^n vectors.*

The key insight is that computing π_1^{tan} reduces, in our finite model, to computing the fiber functor ω —a polynomial-time operation. The Tannakian group isomorphism check is delegated to equality of semantic vectors, which is linear time.

8 Lean 4 Formalization

The companion file `Paper15.TannakianReconstruction.lean` formalizes the core results in LEAN 4. The formalization is organized as follows:

1. **Basic types** (§§ 1–3): `ProgramFragment`, `SemanticVector`, and `FiberFunctor` structures with `DecidableEq` instances.
2. **Fiber functor properties** (§§ 4–6): The semantic equivalence relation `semanticallyEquiv` is proved to be an equivalence relation (reflexivity, symmetry, transitivity), and the tensor product dimension additivity is established.
3. **Verified morphisms** (§§ 5–6): Identity morphisms are verified, and the composition of verified morphisms is verified.
4. **Tannakian group** (§§ 7): The type `piTan` of fiber automorphisms is defined, with composition proved to form a monoid (identity, associativity, unit laws).

5. **Reconstruction** (§ 8): The core reconstruction principle is stated as an axiom `tannakian_reconstruction` (its proof requires the full theory of affine group schemes).
6. **Refactoring validation** (§§ 9–10): The `checkRefactoring` decision procedure is defined and proved sound and complete (`checkRefactoring_sound`, `checkRefactoring_complete`).
7. **Equivalence criterion** (§§ 11–12): The main theorem `tannakian_equiv_criterion` and the refactoring correctness theorem `validation_correct` are proved.

The formalization uses no `sorry` calls; the one axiom (`tannakian_reconstruction`) corresponds to the classical Grothendieck–Deligne–Milne theorem, which is beyond the scope of the present formalization but is stated precisely.

9 Related Work

Tannakian categories in mathematics. The Tannakian formalism was introduced by Grothendieck [5] in the context of motives and systematized by Deligne and Milne [3] and Deligne [2]. Joyal and Street [7] gave a categorical reformulation using braided monoidal categories. The connection to Hopf algebras and quantum groups is developed in [14].

Program equivalence. Semantic equivalence of programs has been studied from many angles: bisimulation [8], logical relations [10], denotational semantics [12], and more recently, translation validation [11] and program synthesis [13]. The present work adds a group-theoretic dimension: the Tannakian group is a new invariant of program equivalence.

Categorical program semantics. The use of category theory in programming language semantics is well-established [9, 6]. Closest to our work is the use of monoidal categories for parallel composition [1] and the categorical semantics of linear logic [4]. Our contribution is to bring *Tannakian duality*—a tool from arithmetic geometry—into this setting.

Earlier JuGeo papers. The sheaf-theoretic foundation (Paper 1), descent obstructions (Paper 3), trust algebra (Paper 4), and characteristic classes (Paper 17) all provide infrastructure used in the present paper. The Tannakian reconstruction theorem is a natural complement to the nerve theorem of Paper 11 and the characteristic class theory of Paper 17: all three extract global invariants from the local structure of verified program judgments.

10 Conclusion

We have developed a Tannakian reconstruction theory for verified program semantics. The category **Prog** of JU GEO verified program fragments admits a neutral Tannakian structure, and the Tannakian reconstruction theorem identifies the group of semantics-preserving transformations as $\mathcal{G}^{\text{tan}} = \underline{\text{Aut}}^{\otimes}(\omega)$.

The Tannakian fundamental group $\pi_1^{\text{tan}}(\mathbf{Prog}, P)$ is a new invariant of programs that captures their semantic symmetry. Two programs are equivalent iff they have isomorphic Tannakian groups (Theorem 6.4), and the TANCHECK algorithm exploits this to validate refactorings in linear time (Theorems 7.2 and 7.3).

Future directions.

- **Infinite programs:** extend π_1^{tan} to programs with infinitely many behaviors using pro-finite group schemes.
- **Non-semisimple case:** study the derived Tannakian formalism for programs with non-trivial obstructions (Paper 3 cohomology classes).
- **Motivic integration:** relate $\pi_1^{\text{tan}}(\mathbf{Prog})$ to the motivic measure of Paper 30, interpreting the Tannakian group as a motivic Galois group.
- **Machine learning:** apply π_1^{tan} to neural network weight-space symmetries, where the Tannakian group captures the symmetry group of the loss landscape.

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